

Sahil Chukka

sahil.chukka@gmail.com — +91 9326609052 — Portfolio — LinkedIn — GitHub

Professional Summary

AI/ML Engineer and Backend Developer with 1+ year of experience shipping production applications powered by machine learning and GenAI. Hands-on with computer vision (CNN classification/detection), regression modeling (XGBoost, Optuna), and Retrieval Augmented Generation (LangChain, ChromaDB), delivered via FastAPI/Flask on AWS and Azure.

Technical Skills

AI / ML: PyTorch, Keras, Scikit-learn, XGBoost, Optuna (Bayesian tuning), CNN Architectures (VGG16, ResNet50, DenseNet121, InceptionV3, EfficientNet), Computer Vision

GenAI: LLMs, RAG, LangChain, CrewAI, Phidata, Hugging Face, Vector Embeddings, ChromaDB

Programming & Backend: Python, TypeScript, JavaScript, SQL, FastAPI, Flask, React, REST APIs

Data & Cloud: AWS, Microsoft Azure, MySQL, MSSQL Server, Power BI, Git

Other: Model Deployment, Offline-First Architecture, PWA, Postman

Professional Experience

TPF Engineering Pvt Ltd, Navi Mumbai

Assistant Manager – BI (Development)

Sept 2025 – Present

Promoted from BI Executive

- Own end-to-end development of **MBIU Pro**, a full-stack PWA (React, TypeScript, FastAPI, SQL Server) for bridge/culvert inspections; integrated a **CNN-based computer vision inference API** for automated crack detection and severity scoring, cutting inspection reporting time from ~15 days to a few hours.
- Architected offline-first data capture with IndexedDB and bi-directional sync for inspection data, images, and GPS coordinates across low-connectivity field sites; built FastAPI services for CRUD, Microsoft SSO auth, image uploads, and automated PDF report generation.
- Built a **RAG-based chatbot** using **Gemini 2.5 Flash** to answer engineer queries on road conditions and inspection doubts, grounding responses in inspection records and domain documentation to speed up field decision-making.
- Automated highway EIRR/cost analysis (**EIRRix Console**, React/Next.js + FastAPI + SQL Server, IRC SP 30 formulas), cutting analysis time from 3–4 days to 1–2 hours.

Data Science Engineer Intern

Apr 2025 – Sept 2025

Finisar Technology India Pvt Ltd, Hyderabad

- Built agent-based CAD dimension-extraction and query-processing services (Flask APIs) with Base64 image parsing and structured data extraction, deployed on AWS for automated CAD file generation.
- Developed an **XGBoost regression pipeline** (skewness correction, feature scaling, Optuna Bayesian tuning) to predict Mn concentration in Garnet from geochemical data; evaluated via MAE and deployed on AWS.
- Built **SalesBot**, a RAG-based conversational AI indexing product manuals into **ChromaDB** with vector embeddings for semantic search and context-aware responses, exposed via Flask APIs on AWS.

Projects

ultFMT – Developer Utilities for the AI Era (ultfmt.com)

June 2026

- Solo-built and shipped a suite of privacy-first web tools for AI engineers: **MCP Studio** (Visual Builder to construct/export Model Context Protocol server configs without hand-writing JSON; MCP Config Validator with schema validation and health scoring), **ML Studio** (Feature Intelligence computing Pearson/Spearman correlation, VIF, PCA, Mutual Information, and ANOVA F-scores; Dataset Health; Learning Curve Plotter), a **RAG Playground**, an **LLM Cost Analysis** calculator, and prompt utilities (Token Estimator, Prompt Diff, Context Window Checker).

Detection of Choroidal Neovascularization (CNV) Using Deep Learning

May 2023 – May 2024

- Benchmarked five CNN architectures – **VGG16**, **ResNet50**, **DenseNet121**, **InceptionV3**, **EfficientNet** – for CNV detection on OCT scans, achieving **95%+ classification accuracy** and strong ROC-AUC.
- Ran comparative analysis of accuracy, computation time, and model complexity, and deployed the best-performing model as a Flask web app for real-time OCT scan upload and automated CNV prediction.

Achievements & Certifications

- Published research paper on CNV detection using deep learning (IJIGSP, DOI: 10.5815/ijigsp.2025.04.07, Aug 2025); application copyright-registered by Govt. of India (ROC No. L-147798/2024)
- Microsoft Certified: Azure AI Fundamentals (AI-900), Aug 2025 — Generative AI course with LangChain and Hugging Face, May 2025

Education

Bachelor of Technology in Information Technology

May 2020 – May 2024

University of Mumbai (Pillai College of Engineering, New Panvel) — **CGPA: 8.22 / 10**